Iván Sevillano-García

521 citations | h-index 4 | 512 since 2021

DaSCI Institute, University of Granada, Spain

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Profile

Post-doc Researcher in Trustworthy AI — explainable AI (XAI), out-of-distribution detection, explanation evaluation & auditing, and explanation-driven model improvement. PhD Cum Laude (UGR, 2025); published in Pattern Recognition, Machine Intelligence Research, Information Fusion, and IEEE JBHI. Current research interests include agentic AI, harness engineering, and loop engineering.

Education

PhD, Information & Communication Technologies — 2021–2025

University of Granada — DaSCI Institute

Thesis: Explainable AI: Methods for the Generation, Evaluation, and Correction of Explanations

Supervisors: F. Herrera, J. Luengo Defense: Nov 13, 2025 **Cum Laude**

Contributions: REVEL (explanation evaluation), X-SHIELD (explanation regularization), STOOD-X (explainable OOD detection). hdl.handle.net/10481/112935

ISBN 9791370150433

M.Sc. Data Science & Computer Engineering — 2019–2020

University of Granada

Degree in Mathematics — 2013–2019

University of Granada — ETSIIT

Degree in Computer Science — 2013–2019

University of Granada — ETSIIT

Research Experience

Predoctoral Researcher — 2019–2025

University of Granada — DaSCI Institute

- **REVEL:** designed a theoretical framework to audit local linear explanations for robustness, conciseness, and prescriptivity; evaluated on deep learning image classification benchmarks.
- **X-SHIELD:** proposed an explanation-driven regularization family that selects input features to hide during training, improving model performance and explainability simultaneously.
- **STOOD-X:** developed a two-stage explainable OOD detection method combining Wilcoxon-Mann-Whitney nonparametric testing with concept-based visual explanations.
- **COVIDGR:** co-designed a balanced COVID-19 chest X-ray dataset across all severity levels and the COVID-SDNet methodology.

Visiting Researcher — Feb–Mar 2026

Poznan University of Technology, Poland — host: Prof. J. Stefanowski

- Research on prototype-based and counterfactual explanations for XAI post-hoc methods.

Industrial Collaboration — ArcelorMittal — 2020–2021

via DaSCI, UGR

- Co-developed a deep-learning ensemble for metallographic image segmentation; co-released the MetalDAM dataset.

Publications

Journals

STOOD-X: Explainable OOD Detection via Nonparametric Statistical Testing on Large-Scale Datasets.

Pattern Recognition, 2026. **I. Sevillano-García**, J. Luengo, F. Herrera. DOI (3 cit.)

X-SHIELD: Regularization for Explainable Artificial Intelligence.

Machine Intelligence Research, 23, 528–544, 2026. **I. Sevillano-García**, J. Luengo, F. Herrera. DOI (9 cit.)

A Three-Level Framework for LLM-Enhanced eXplainable AI.

Information Systems Frontiers, 2025. M. Bello, R. Bello, M.M. García, A. Nowé, **I. Sevillano-García**, F. Herrera. DOI (14 cit.)

REVEL Framework to Measure Local Linear Explanations for Black-Box Models.

Intl. J. Intelligent Systems, 2023. **I. Sevillano-García**, J. Luengo, F. Herrera. DOI (15 cit.)

A Tutorial on Segmentation of Metallographic Images: MetalDAM Dataset, DL Ensemble.

Information Fusion, 2022. J. Luengo, R. Moreno, **I. Sevillano-García**, et al. DOI (59 cit.)

COVIDGR Dataset & COVID-SDNet Methodology for Predicting COVID-19 (Chest X-ray).

IEEE JBHI, 24(12), 3595–3605, 2020. S. Tabik, A. Gómez-Ríos, J.L. Martín-Rodríguez, **I. Sevillano-García**, et al. DOI (477 cit.)

Conferences

Fine-Grained Prostate Cancer Tissue Classification via Expert-Annotated Dataset & Interpretable DL.

IEEE CAI 2026, 1264–1269. J. Ayensa-Jiménez, D. Navarro, **I. Sevillano-García**, et al. DOI

Optimizing LIME Explanations Using REVEL Metrics.

HAIS, 304–313, 2023. **I. Sevillano-Garcia**, J. Luengo, F. Herrera. DOI

Low-Impact Feature Reduction Regularization Term: How to Improve AI with Explainability.

xAI 2023 (Late-breaking Work, poster), 135–139. **I. Sevillano-García**, J. Luengo, F. Herrera. DBLP

Slicer: Feature Learning for Class Separability with LS-SVM Loss (COVID-19 X-ray).

HAIS, 305–315, 2021. D. Charte, **I. Sevillano-García**, et al. DOI

Open-Source Software & Datasets

Software

- **S-ReVel** — audits local linear explanations (e.g. LIME) for robustness, conciseness, and prescriptivity. github.com/ari-dasci/S-ReVel. AGPL-3.0.
- **S-STOOD-X** — explainable OOD detection via Wilcoxon-Mann-Whitney testing and concept-based visual explanations. github.com/ari-dasci/S-STOOD-X. GPL-3.0.
- **S-T-SHIELD** — explanation-driven regularization that hides input features during training to improve performance and explainability. github.com/ari-dasci/S-T-SHIELD. GPL-3.0.

Datasets

- **COVIDGR** — COVID-19 chest X-ray dataset with fine-grained classification across severity levels, from positive RT-PCR with no symptoms to severe cases. github.com/ari-dasci/OD-covidgr.
- **MetalDAM** — metallographic image dataset for microstructural segmentation in steel industry quality control. github.com/ari-dasci/OD-MetalDAM.

Teaching & Mentoring

- Co-supervision of 2 PhD students (UGR program), ongoing: topics include Machine Unlearning, XAI metrics, and Agentic AI.
- Co-supervision of undergraduate and master's theses, UGR.
- University teaching: Programming Methodology (1st-year undergraduate course, Dept. of CS & AI, UGR, 2024–2025). Course page.

Academic Service

Invited Talks & Teaching:

- ADIA Lab International Summer School on Explainable AI — "Lab on xAI Tools" (Granada, May 2025).
- Instructor, INCIBE Seminars on AI & Cybersecurity (DaSCI / Chamber of Commerce, 2025). Coordinated Block 1 on Trustworthy AI with F. Herrera.

Peer Review:

- **Information Sciences** (Elsevier; IF 6.0) — 2025. Information science, knowledge engineering, intelligent systems.
- **Information Fusion** (Elsevier; IF 17.4) — 2023. Multi-sensor, multi-source information fusion.
- **IEEE SSCI 2025** — Symposium on Trustworthy, Explainable and Responsible Computational Intelligence (Trondheim, Norway).

Verified via ORCID and official CMT confirmation.

Research Projects & Funding

- **PID2023-150070NB-I00** (MCIN/AEI). *Predoctoral and postdoctoral researcher*. Trustworthy AI, focused on explainability and safe AI.
- **ENIA IAFER Chair** (EU NextGen, TSI-100927-2023-1). *Predoctoral researcher*. Trustworthy AI, explainability and safe AI.
- **IAFER CiB — INCIBE Chair on Ethical AI for Cybersecurity** (DaSCI/UGR). *Predoctoral researcher*. XAI applied to cybersecurity and ethical AI.

Technical Skills

Languages: Python, R
Frameworks: PyTorch, TensorFlow, scikit-learn
Tools: Git, Linux, Conda, TensorBoard, Slurm
Computing: multi-GPU clusters (NVIDIA A100), distributed training, experiment tracking

Spoken Languages

Spanish — native | English — fluent

References

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Prof. Julián Luengo
Full Professor
Dept. CS & AI, UGR
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